Optimization

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```
# The Python chunks on this page import these; reticulate provisions them.
# Attaching reticulate gives the inline R its `py` object.
library(reticulate)
source("../tools/use-torch.R")
use_torch(c("numpy", "pandas", "matplotlib"))
```

Brian Hutchinson's CSCI 481/581 sets the order of topics in these notes, which follow it with his permission; this page covers the training algorithms and starting weights of his module on optimization and regularization[1]. The exercises are new. Minibatches, momentum, Adam and starting weights follow *Deep Learning*, Chapter 8 [1].

1 Welcome

1.1 Plan for this lecture

The feedforward networks chapter trained its MLP with full-batch gradient descent[2], and lds's chapter on gradient descent[3] covers the algorithms that do better. This lecture runs them in PyTorch on the digits:

1. training on small random subsets of the training images instead of all of them at each step;
2. momentum[4] and Adam[5] as PyTorch optimizers;
3. changing the learning rate[6] during training, starting with a short ramp up;
4. choosing the scale of the starting weights.

1.2 The digits and the network

```
import math

import numpy as np
import pandas as pd
import torch
from torch import nn
import matplotlib.pyplot as plt

_ = torch.manual_seed(481)
```

[1] https://morrison-lab.github.io/lds/chapters/big-ideas.html#def-regularization
[2] https://morrison-lab.github.io/mds/optimization.html#def-gradient-descent
[3] https://morrison-lab.github.io/lds/chapters/gradient-descent.html
[4] https://morrison-lab.github.io/lds/chapters/gradient-descent.html#def-momentum
[5] https://morrison-lab.github.io/lds/chapters/gradient-descent.html#def-adam
[6] https://morrison-lab.github.io/lds/chapters/gradient-descent.html#def-learning-rate

```
digits = pd.read_csv("data/optdigits-test.csv", header=None)
x = torch.tensor(digits.iloc[:, :64].to_numpy(), dtype=torch.float32) / 16
y = torch.tensor(digits[64].to_numpy())
n_train = 1200
order = torch.randperm(len(x))
x_train, y_train = x[order[:n_train]], y[order[:n_train]]
x_test, y_test = x[order[n_train:]], y[order[n_train:]]
loss_fn = nn.CrossEntropyLoss()


def make_mlp(seed=1):
    # A fresh copy of the digits MLP, with the same starting weights for a given seed.
    torch.manual_seed(seed)
    return nn.Sequential(nn.Linear(64, 32), nn.ReLU(), nn.Linear(32, 10))


print(make_mlp())
```

```
Sequential(
  (0): Linear(in_features=64, out_features=32, bias=True)
  (1): ReLU()
  (2): Linear(in_features=32, out_features=10, bias=True)
)
```

The exercises use the 1,797 digits of `data/optdigits-test.csv`[7] [2], split as in the feedforward networks chapter, with 1,200 images in the training set[8]. `make_mlp()` builds that chapter's MLP, with the same starting weights each time, so that every method below starts from the same place.

2 Minibatches

2.1 Ten passes over the data

Exercise 2.1 (Ten passes over the data)**.** Suppose one fresh MLP from `make_mlp()` is trained for 10 epochs[a] of full-batch gradient descent[b] with learning rate[c] 0.5. Another is trained for 10 epochs of minibatch gradient descent[d] with batch size 32 and learning rate 0.1. The minibatches of each pass come from a fresh shuffle of the training images, rather than being drawn with replacement. Which has the lower training loss after 10 epochs?

[a] https://morrison-lab.github.io/lds/chapters/gradient-descent.html#def-epoch
[b] https://morrison-lab.github.io/mds/optimization.html#def-gradient-descent
[c] https://morrison-lab.github.io/lds/chapters/gradient-descent.html#def-learning-rate
[d] https://morrison-lab.github.io/lds/chapters/gradient-descent.html#def-minibatch

Solution

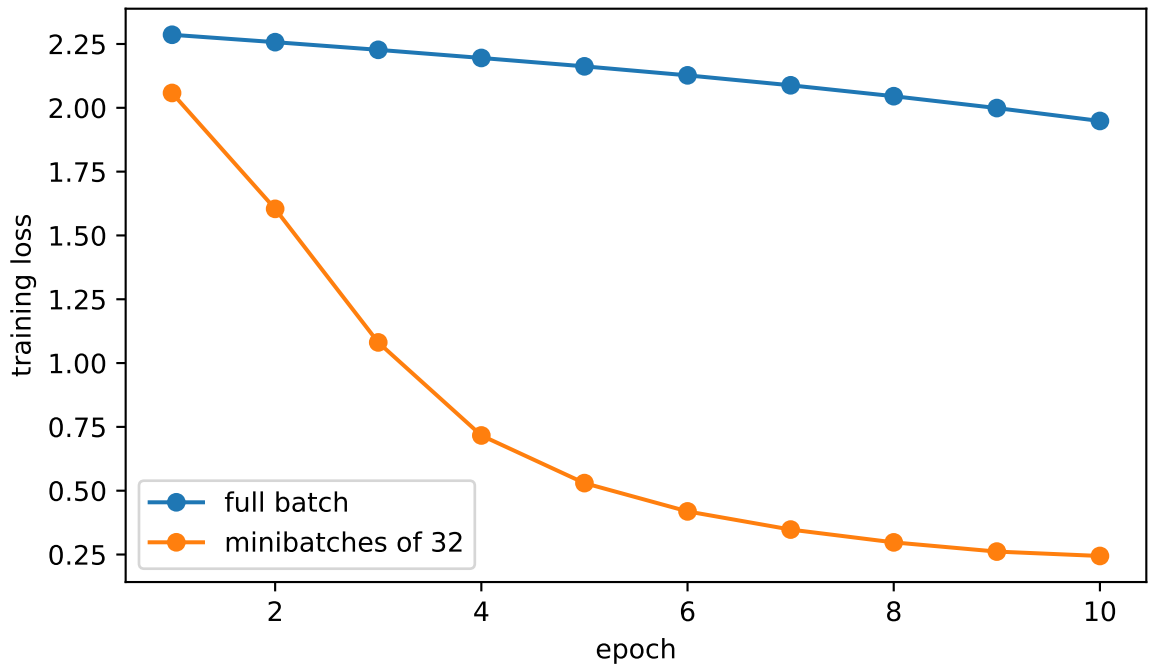
Solution 2.1.

[7] https://morrison-lab.github.io/dds/chapters/data/optdigits-test.csv
[8] https://morrison-lab.github.io/lds/chapters/big-ideas.html#def-training-set

```
def train(mlp, optimizer, n_epochs, batch_size, scheduler=None, seed=0):
    # Minibatch training; returns the training loss after each epoch.
    shuffler = torch.Generator().manual_seed(seed)
    losses = []
    for epoch in range(n_epochs):
        shuffled = torch.randperm(n_train, generator=shuffler)
        for start in range(0, n_train, batch_size):
            batch = shuffled[start:start + batch_size]
            optimizer.zero_grad()
            loss_fn(mlp(x_train[batch]), y_train[batch]).backward()
            optimizer.step()
            if scheduler is not None:
                scheduler.step()
        with torch.no_grad():
            losses.append(loss_fn(mlp(x_train), y_train).item())
    return losses


full_mlp = make_mlp()
full_losses = train(full_mlp, torch.optim.SGD(full_mlp.parameters(), lr=0.5), 10, n_train)
mini_mlp = make_mlp()
mini_losses = train(mini_mlp, torch.optim.SGD(mini_mlp.parameters(), lr=0.1), 10, 32)

fig, ax = plt.subplots(figsize=(6, 3.5), layout="constrained")
ax.plot(range(1, 11), full_losses, marker="o", label="full batch");
ax.plot(range(1, 11), mini_losses, marker="o", label="minibatches of 32");
ax.set_xlabel("epoch");
ax.set_ylabel("training loss");
ax.legend();
plt.show()
```



Minibatches win by far: after 10 epoch[a]s the training loss is 0.24 for minibatches of 32 and 1.95 for the full batch. An epoch costs about the same either way, since it computes the loss and its gradient[b] once for each training image. Full-batch descent spends that cost on one step per epoch,

while the minibatch version spends it on many smaller, noisier steps, each of which already points roughly downhill. `batch_size=n_train` makes `train()` take full-batch steps.

[a] https://morrison-lab.github.io/lds/chapters/gradient-descent.html#def-epoch
[b] https://morrison-lab.github.io/lds/chapters/linear-regression.html#def-gradient

2.2 How noisy is a minibatch gradient?

Exercise 2.2 (How noisy is a minibatch gradient?)**.** lds's exercise on minibatches[a] shows that, for minibatches drawn with replacement, the minibatch gradient[b]'s variance is $1/b$ times that of one image's gradient, for batch size b. For a fresh MLP at its starting parameters[c], and 200 minibatches each of sizes 8, 32 and 128, does b times the mean squared distance between the minibatch gradient and the full-batch gradient stay about the same?

[a] https://morrison-lab.github.io/lds/chapters/gradient-descent.html#exr-minibatch
[b] https://morrison-lab.github.io/lds/chapters/linear-regression.html#def-gradient
[c] https://morrison-lab.github.io/lds/chapters/big-ideas.html#def-parameter

Solution

Solution 2.2.

```
noise_mlp = make_mlp()


def flat_gradient(rows):
    # The loss gradient on the given training images, as one long vector.
    noise_mlp.zero_grad()
    loss_fn(noise_mlp(x_train[rows]), y_train[rows]).backward()
    return torch.cat([p.grad.flatten() for p in noise_mlp.parameters()])


full_gradient = flat_gradient(torch.arange(n_train))
sampler = torch.Generator().manual_seed(0)
noise = {}
for b in [8, 32, 128]:
    distances = []
    for draw in range(200):
        rows = torch.randint(n_train, (b,), generator=sampler)
        distances.append(((flat_gradient(rows) - full_gradient) ** 2).sum().item())
    noise[b] = np.mean(distances)
print({b: round(float(b * d), 2) for b, d in noise.items()})
```

```
{8: 4.55, 32: 4.6, 128: 4.44}
```

Yes. The mean squared distance falls from 0.569 at $b = 8$ to 0.035 at $b = 128$, and b times it stays between 4.4 and 4.6. The mean squared distance is the sum of the variances of the gradient[a]'s entries, so it falls as $1/b$, as each variance does. Quadrupling the batch size only halves the typical distance, which is one reason small batches are a good use of each gradient computation.

[a] https://morrison-lab.github.io/lds/chapters/linear-regression.html#def-gradient

3 Momentum and Adam

3.1 Momentum in PyTorch

Remark (Momentum in PyTorch)

Remark 3.1 (Momentum in PyTorch). `torch.optim.SGD(params, lr=alpha, momentum=beta)` takes the steps of lds's momentum[a], $\underset{\sim}{v}_t = \beta \underset{\sim}{v}_{t-1} + \underset{\sim}{g}_t$ and $\vec{\beta}_t = \vec{\beta}_{t-1} - \alpha \underset{\sim}{v}_t$, where $\underset{\sim}{g}_t$ is the gradient[b] at step t and $\vec{\beta}_t$ the parameters[c] after it. Its first step sets $\underset{\sim}{v}_1 = \underset{\sim}{g}_1$, as $\underset{\sim}{v}_0 = \vec{0}$ would [3]. Some sources scale the gradient term by $1-\beta$ or fold α into $\underset{\sim}{v}_t$; the steps are the same up to a rescaled learning rate[d].

[a] https://morrison-lab.github.io/lds/chapters/gradient-descent.html#def-momentum
[b] https://morrison-lab.github.io/lds/chapters/linear-regression.html#def-gradient
[c] https://morrison-lab.github.io/lds/chapters/big-ideas.html#def-parameter
[d] https://morrison-lab.github.io/lds/chapters/gradient-descent.html#def-learning-rate

3.2 Momentum on a steady slope

Exercise 3.1 (Momentum on a steady slope)**.** Suppose the gradient[a] is the same vector g at every step, and momentum[b], updated as in Remark 3.1, starts from $\underset{\sim}{v}_0 = \vec{0}$. What does the step $\alpha \underset{\sim}{v}_t$ approach as t grows?

[a] https://morrison-lab.github.io/lds/chapters/linear-regression.html#def-gradient
[b] https://morrison-lab.github.io/lds/chapters/gradient-descent.html#def-momentum

Solution

Solution 3.1. Unrolling $\underset{\sim}{v}_t = \beta \underset{\sim}{v}_{t-1} + \underset{\sim}{g}$ from $\underset{\sim}{v}_0 = \vec{0}$:

$$\begin{aligned} \underset{\sim}{v}_t &= \underset{\sim}{g} + \beta \underset{\sim}{g} + \cdots + \beta^{t-1} \underset{\sim}{g} && \text{(apply the update } t \text{ times)} \\ &= \frac{1-\beta^t}{1-\beta} \underset{\sim}{g} && \text{(sum of a geometric series, for } 0 \le \beta < 1) \\ &\to \frac{1}{1-\beta} \underset{\sim}{g} && \text{(since } \beta^t \to 0) \end{aligned}$$

So the step approaches $\frac{\alpha}{1-\beta} \underset{\sim}{g}$: on a steady slope, momentum[a] moves like plain gradient descent[b] with learning rate[c] $\alpha/(1-\beta)$. With $\beta = 0.9$, that learning rate is 10 times α.

[a] https://morrison-lab.github.io/lds/chapters/gradient-descent.html#def-momentum
[b] https://morrison-lab.github.io/mds/optimization.html#def-gradient-descent
[c] https://morrison-lab.github.io/lds/chapters/gradient-descent.html#def-learning-rate

3.3 Three optimizers on the digits

Exercise 3.2 (Plain steps, momentum and Adam)**.** Suppose three fresh MLPs are trained for 5 epoch[a]s on minibatches of 32, using `train()` from Solution 2.1:

- one with `torch.optim.SGD` at learning rate[b] 0.1;
- one with `torch.optim.SGD` at learning rate 0.01 and momentum[c] 0.9;
- one with `torch.optim.Adam`, PyTorch's Adam[d], at learning rate 0.01.

Which reaches the lowest training loss?

[a] https://morrison-lab.github.io/lds/chapters/gradient-descent.html#def-epoch

[b] https://morrison-lab.github.io/lds/chapters/gradient-descent.html#def-learning-rate
[c] https://morrison-lab.github.io/lds/chapters/gradient-descent.html#def-momentum
[d] https://morrison-lab.github.io/lds/chapters/gradient-descent.html#def-adam

Solution

Solution 3.2.

```
optimizers = {
    "SGD": lambda p: torch.optim.SGD(p, lr=0.1),
    "momentum": lambda p: torch.optim.SGD(p, lr=0.01, momentum=0.9),
    "Adam": lambda p: torch.optim.Adam(p, lr=0.01),
}
final_loss, test_acc = {}, {}
for name, make_optimizer in optimizers.items():
    mlp = make_mlp()
    final_loss[name] = train(mlp, make_optimizer(mlp.parameters()), 5, 32)[-1]
    with torch.no_grad():
        test_acc[name] = (mlp(x_test).argmax(dim=1) == y_test).float().mean().item()
print(final_loss)
```

```
{'SGD': 0.5297544598579407, 'momentum': 0.5874408483505249, 'Adam': 0.10584728419780731}
```

```
print(test_acc)
```

```
{'SGD': 0.8860971331596375, 'momentum': 0.8844221234321594, 'Adam': 0.94807368516922}
```

Adam[a] reaches the lowest training loss, 0.11, against 0.53 for plain SGD and 0.59 with momentum[b]. It also classifies the most images of the test set[c] correctly, 94.8%. Momentum and plain SGD end close together, as Solution 3.1 predicts: learning rate[d] 0.01 with $\beta = 0.9$ moves like learning rate 0.1 on a steady slope. A comparison like this depends on the learning rates chosen for each method, so a fair one tunes each method's learning rate separately.

[a] https://morrison-lab.github.io/lds/chapters/gradient-descent.html#def-adam
[b] https://morrison-lab.github.io/lds/chapters/gradient-descent.html#def-momentum
[c] https://morrison-lab.github.io/lds/chapters/big-ideas.html#def-test-set
[d] https://morrison-lab.github.io/lds/chapters/gradient-descent.html#def-learning-rate

3.4 Adam's first step

Exercise 3.3 (Adam's first step)**.** At Adam[a]'s first step, one parameter[b] has gradient[c] $g \neq 0$. Ignoring ε, how far does that step move the parameter?

[a] https://morrison-lab.github.io/lds/chapters/gradient-descent.html#def-adam
[b] https://morrison-lab.github.io/lds/chapters/big-ideas.html#def-parameter
[c] https://morrison-lab.github.io/lds/chapters/linear-regression.html#def-gradient

Solution

Solution 3.3. With $m_0 = u_0 = 0$, the first averages are $m_1 = (1 - \beta_1)g$ and $u_1 = (1 - \beta_2)g^2$.

$$\begin{aligned}\frac{\hat{m}_1}{\sqrt{\hat{u}_1}} &= \frac{m_1/(1-\beta_1)}{\sqrt{u_1/(1-\beta_2)}} && \text{(bias corrections at } t=1\text{)}\\ &= \frac{g}{\sqrt{g^2}} && \text{(substitute } m_1 \text{ and } u_1\text{)}\\ &= \frac{g}{|g|} && (\sqrt{g^2} = |g|)\end{aligned}$$

So the first step moves the parameter[a] by exactly α, against the sign of g, whatever the size of g.

```
w = torch.ones(2, requires_grad=True)
adam = torch.optim.Adam([w], lr=0.01)
w.grad = torch.tensor([1000.0, 0.001])
adam.step()
step = (1 - w.detach()).tolist()
print(step)
```

```
[0.009999990463256836, 0.009999871253967285]
```

Gradients[b] of 1,000 and 0.001 move their parameters[c] by 0.0100000 and 0.0099999, both close to the learning rate[d], 0.01. The small difference is the effect of ε. So the learning rate of Adam[e] sets the size of each step on the parameters[f]' own scale. Later steps are also bounded by about the learning rate [4, Sec. 2.1]. A step of plain gradient descent[g] is instead the learning rate times the gradient[h].

[a] https://morrison-lab.github.io/lds/chapters/big-ideas.html#def-parameter
[b] https://morrison-lab.github.io/lds/chapters/linear-regression.html#def-gradient
[c] https://morrison-lab.github.io/lds/chapters/big-ideas.html#def-parameter
[d] https://morrison-lab.github.io/lds/chapters/gradient-descent.html#def-learning-rate
[e] https://morrison-lab.github.io/lds/chapters/gradient-descent.html#def-adam
[f] https://morrison-lab.github.io/lds/chapters/big-ideas.html#def-parameter
[g] https://morrison-lab.github.io/mds/optimization.html#def-gradient-descent
[h] https://morrison-lab.github.io/lds/chapters/linear-regression.html#def-gradient

4 Changing the learning rate

4.1 Starting slowly

Definition 4.1 (Learning-rate warmup)**. Learning-rate warmup** raises the learning rate[a] from a small value to its target over the first steps of training, usually linearly, before the rest of the schedule takes over [5, Sec. 2.2]. With linear warmup over T_w steps to a target learning rate α, step $t = 0, 1, \ldots, T_w - 1$ uses learning rate $\alpha(t+1)/T_w$.

[a] https://morrison-lab.github.io/lds/chapters/gradient-descent.html#def-learning-rate

Example 4.1 (Warmup to 0.1 over 100 steps)**.**

```
target_rate, n_warmup_steps = 0.1, 100
warmup_rates = {t: target_rate * (t + 1) / n_warmup_steps for t in [0, 49, 99]}
print(warmup_rates)
```

```
{0: 0.001, 49: 0.05, 99: 0.1}
```

With $T_w = 100$ and $\alpha = 0.1$, learning-rate warmup uses learning rate[a] 0.001 at step 0, 0.05 at step 49, and 0.1 at step 99, the last step of the warmup.

[a] https://morrison-lab.github.io/lds/chapters/gradient-descent.html#def-learning-rate

4.2 A schedule in PyTorch

Exercise 4.1 (Warmup, then cosine decay)**.** A schedule for `torch.optim.SGD` at learning rate[a] 0.1 uses learning-rate warmup over 100 steps, and then cosine decay[b] to 0 at step 1,000. What learning rate does it give at step 550?

[a] https://morrison-lab.github.io/lds/chapters/gradient-descent.html#def-learning-rate
[b] https://morrison-lab.github.io/lds/chapters/gradient-descent.html#def-cosine-decay

Solution

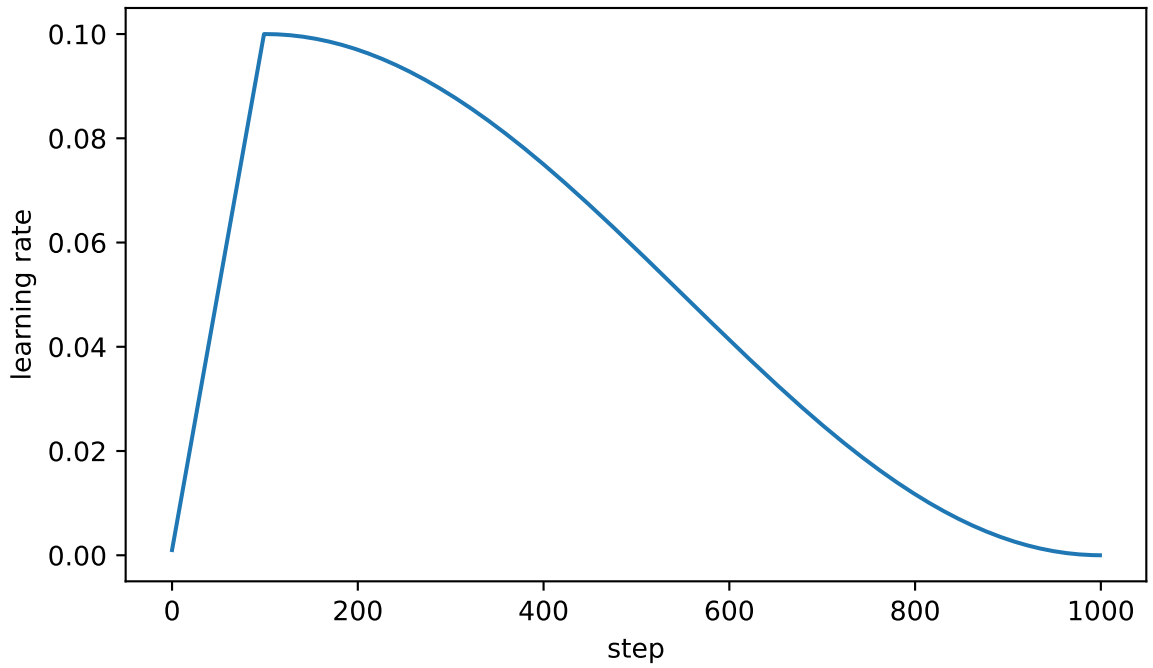
Solution 4.1.

```
n_warmup, n_steps = 100, 1000


def warmup_cosine(step):
    # The factor that multiplies the base learning rate at this step.
    if step < n_warmup:
        return (step + 1) / n_warmup
    progress = (step - n_warmup) / (n_steps - n_warmup)
    return 0.5 * (1 + math.cos(math.pi * progress))


schedule_mlp = make_mlp()
sgd = torch.optim.SGD(schedule_mlp.parameters(), lr=0.1)
scheduler = torch.optim.lr_scheduler.LambdaLR(sgd, warmup_cosine)
rates = []
for step in range(n_steps):
    rates.append(sgd.param_groups[0]["lr"])
    sgd.step()
    scheduler.step()

fig, ax = plt.subplots(figsize=(6, 3.5), layout="constrained")
ax.plot(rates);
ax.set_xlabel("step");
ax.set_ylabel("learning rate");
plt.show()
```




```
rate_550 = rates[550]
```

0.050. Step 550 is halfway through the cosine part, from step 100 to step 1,000, and halfway there cosine decay[a] gives half the base learning rate[b], since $\frac{1}{2}(1+\cos(\frac{\pi}{2}))=\frac{1}{2}$. The loop calls `scheduler.step()` after `sgd.step()`, the order PyTorch expects, and `train()` in Solution 2.1 does the same at every minibatch. Here `sgd.step()` changes nothing, since no gradients[c] have been computed; it is there only to show the order.

[a] https://morrison-lab.github.io/lds/chapters/gradient-descent.html#def-cosine-decay
[b] https://morrison-lab.github.io/lds/chapters/gradient-descent.html#def-learning-rate
[c] https://morrison-lab.github.io/lds/chapters/linear-regression.html#def-gradient

ℹ Remark (Why warm up)

Remark 4.1 (Why warm up). At the start of training the parameters[a] are far from good ones, and the gradients[b] can be large, so a learning rate[c] that works later can throw the network into a bad region in the first few steps. Goyal et al. [5] proposed gradual warmup for training with very large minibatches. For Adam[d], its running average of squared gradients rests on only a few gradients at first, so its step sizes vary widely, and warmup keeps those early steps small [6].

[a] https://morrison-lab.github.io/lds/chapters/big-ideas.html#def-parameter
[b] https://morrison-lab.github.io/lds/chapters/linear-regression.html#def-gradient
[c] https://morrison-lab.github.io/lds/chapters/gradient-descent.html#def-learning-rate
[d] https://morrison-lab.github.io/lds/chapters/gradient-descent.html#def-adam

5 Starting weights

5.1 The scale of the starting weights

Exercise 5.1 (How the starting scale grows or shrinks)**.** Suppose the training images are pushed through 10 layers, each computing $\text{ReLU}(\mathbf{W}\tilde{h})$ with width[a] 256 and no bias. Every weight is drawn independently from a normal distribution with mean 0 and standard deviation σ. How does the standard deviation of the last layer's values depend on σ, for $\sigma = 0.01$, for $\sigma = \sqrt{2/n_{\text{in}}}$, where n_{in} is the layer's number of inputs, and for $\sigma = 0.2$?

[a] https://morrison-lab.github.io/lds/chapters/neural-nets.html#def-width

Solution

Solution 5.1.

```
def last_layer_sd(scale, depth=10, width=256):
    # scale(n_in) gives the standard deviation of the weights of a layer with n_in inputs.
    torch.manual_seed(0)
    h = x_train
    for layer in range(depth):
        n_in = h.shape[1]
        weight = scale(n_in) * torch.randn(width, n_in)
        h = torch.relu(h @ weight.T)
    return h.std().item()


init_sd = {
    "small": last_layer_sd(lambda n_in: 0.01),
    "he": last_layer_sd(lambda n_in: math.sqrt(2 / n_in)),
    "large": last_layer_sd(lambda n_in: 0.2),
    "pixels": x_train.std().item(),
}
print(init_sd)
```

```
{'small': 5.173762460120024e-11, 'he': 0.30115261673927307, 'large': 529.7935180664062, 'pixels': 0.37
```

The pixels have standard deviation 0.38. After 10 layers, $\sigma = 0.01$ leaves values with standard deviation about 10^{-10}, $\sigma = 0.2$ gives about 10^3, and $\sigma = \sqrt{2/n_{\text{in}}}$ gives 0.30, close to where it started. Each layer multiplies the typical size of the values by about $\sigma\sqrt{n_{\text{in}}/2}$: the weighted sum of n_{in} values has variance proportional to $n_{\text{in}}\sigma^2$, and the ReLU zeroes about half of the results. That factor is 1 for $\sigma^2 = 2/n_{\text{in}}$, which is He initialization[a] [7]. Since backpropagation multiplies by the same weight matrices on the way back, the gradients[b] shrink or grow layer by layer in the same way.

[a] https://morrison-lab.github.io/lds/chapters/backpropagation.html#def-he-initialization
[b] https://morrison-lab.github.io/lds/chapters/linear-regression.html#def-gradient

5.2 PyTorch's default

Exercise 5.2 (PyTorch's starting weights)**.** `nn.Linear(n_in, n_out)` draws each weight independently and uniformly from the interval $(-1/\sqrt{n_{\text{in}}}, 1/\sqrt{n_{\text{in}}})$ [8]. What is the variance of each weight?

Solution

Solution 5.2. A weight w drawn uniformly from $(-a, a)$ has mean 0, so its variance is

$$\begin{aligned}\operatorname{Var}(w) &= \mathbb{E}[w^2] && \text{(the mean is 0)}\\ &= \int_{-a}^{a} w^2 \frac{1}{2a}\, dw && \text{(the density is } \frac{1}{2a} \text{ on } (-a, a))\\ &= \frac{1}{2a} \cdot \frac{2a^3}{3} && \text{(integrate } w^2 \text{ from } -a \text{ to } a)\\ &= \frac{a^2}{3},\end{aligned}$$

so with $a = 1/\sqrt{n_{\text{in}}}$ each weight has variance $1/(3n_{\text{in}})$.

```
_ = torch.manual_seed(0)
layer = nn.Linear(256, 256)
default_var_times_n = layer.weight.var().item() * 256
print(default_var_times_n)
```

```
0.33143436908721924
```

For `nn.Linear(256, 256)` the sample variance of the weights times 256 is 0.331, close to $\frac{1}{3}$. This variance is $\frac{1}{6}$ of He initialization[a]'s $2/n_{\text{in}}$, so by Solution 5.1 each ReLU layer with PyTorch's starting weights shrinks the values by a factor of about $1/\sqrt{6}$. `nn.init.kaiming_normal_` gives a layer He initialization instead.

[a] https://morrison-lab.github.io/lds/chapters/backpropagation.html#def-he-initialization

5.3 Starting every weight at the same value

Exercise 5.3 (Starting every weight at the same value)**.** Suppose every weight and bias of a fresh MLP is set to 0.1. It is then trained for 5 epoch[a]s on minibatches of 32, with `torch.optim.SGD` at learning rate[b] 0.1. What happens to the 32 units of its hidden layer?

[a] https://morrison-lab.github.io/lds/chapters/gradient-descent.html#def-epoch
[b] https://morrison-lab.github.io/lds/chapters/gradient-descent.html#def-learning-rate

Solution

Solution 5.3.

```
constant_mlp = make_mlp()
with torch.no_grad():
    for p in constant_mlp.parameters():
        _ = p.fill_(0.1)
_ = train(constant_mlp, torch.optim.SGD(constant_mlp.parameters(), lr=0.1), 5, 32)
first_weights = constant_mlp[0].weight
largest_row_gap = (first_weights - first_weights[0]).abs().max().item()
with torch.no_grad():
    constant_acc = (constant_mlp(x_test).argmax(dim=1) == y_test).float().mean().item()
print(largest_row_gap, constant_acc)
```

```
0.0 0.09380234777927399
```

They stay identical: after training, the largest difference between any row of the first layer's weight

matrix and the first row is 0, so all 32 units compute the same function of the pixels, and the network classifies only 9.4% of the test images correctly. Units that start equal receive equal gradients[a], since each enters the output in the same way, so every step changes them equally. The network has, in effect, one hidden unit.

[a] https://morrison-lab.github.io/lds/chapters/linear-regression.html#def-gradient

Remark (Random starting weights break the symmetry)

Remark 5.1 (Random starting weights break the symmetry). Drawing the starting weights at random makes each unit start as a different function, so the units receive different gradients[a] and can come to compute different functions [1, Sec. 8.4]. The biases can start at 0, since the random weights already tell the units apart. Solution 5.1 and Solution 5.2 show how to choose the scale of the random weights.

[a] https://morrison-lab.github.io/lds/chapters/linear-regression.html#def-gradient

5.4 Further reading

Read

Deep Learning's Chapter 8[a] covers minibatches, momentum[b], Adam[c] and starting weights for deep networks [1, Ch. 8].

[a] https://www.deeplearningbook.org/contents/optimization.html
[b] https://morrison-lab.github.io/lds/chapters/gradient-descent.html#def-momentum
[c] https://morrison-lab.github.io/lds/chapters/gradient-descent.html#def-adam

6 References

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